

Wetland_fraction_r2_WIP.R

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NWI_Wetland_fraction	<i>Create gridded wetland coverage maps using National Wetlands Inventory data</i>
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Description

‘NWI_Wetland_fraction’ writes multiple tiff files of the fractional wetland cover in each pixel. There is one file for each state - wetland type combination.

Usage

```
NWI_Wetland_fraction(  
  input_directory,  
  output_directory,  
  domain,  
  domain_template,  
  state_name_list,  
  Use_SOCCR1,  
  Use_SOCCR2,  
  Include_freshwater  
)
```

Arguments

input_directory	Character providing the full filepath to save/load input data
output_directory	Character providing the full filepath to save processed data
domain	SpatVector polygon outlining the desired output area
domain_template	SpatRaster providing the desired output grid, including the desired resolution and coordinate reference system

state_name_list	Character vector listing all states within the desired domain
Use_SOCCR1	Logical. Pulled from config file. Indicating whether or not to calculate emissions using SOCCR1.
Use_SOCCR2	Logical. Pulled from config file. Indicating whether or not to calculate emissions using SOCCR2.
Include_freshwater	Logical. Pulled from config file. Indicating whether or not to calculate emissions for freshwater wetlands using Rosentreter et al.

Details

This function downloads National Wetlands Inventory (NWI) data and calculates the fraction of each gridcell in the domain that is covered by the NWI polygons. This is done separately for each state as the data is available by state or watershed. It also processes each type of wetland separately.

The wetland types are pulled from the NWI system, subsystem, and class as defined here <https://www.fws.gov/media/national-wetland-inventory-wetlands-and-deepwater-map-code-diagram> and discussed in detail here <https://www.fws.gov/media/classification-wetlands-and-deepwater-habitats-> They are M1 (marine, subtidal), M2 (marine, intertidal), E1 (estuarine, subtidal), E2 (estuarine, intertidal), R1 (riverine, tidal), R2 (riverine, lower perennial), R3 (riverine, upper perennial), R4 (riverine, intermittent), L1 (lacustrine, limnetic), L2 (lacustrine, littoral), PFO (palustrine, forested), and PNF (palustrine, all non-forested classes).

The NWI data for each state will be automatically downloaded. This can be up to several GB per state.

Given the high resolution polygons and significant number being processed, this function can be time consuming.

The NWI is a biannually updated dataset including the extent and type of wetlands across the United States. It is available at <https://www.fws.gov/program/national-wetlands-inventory/wetlands-data>.

A separate TIFF file is saved for each state - wetland type combination. These are used in further processing.

Value

Nothing is returned from the function, but the main outputs are TIFF files of the fractional wetland coverage per pixel for each wetland type and state. They are titled "state abbreviation _ wetland type abbreviation.tiff".

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Examples

```
library(terra)
grid_bbox=cbind(c(-76.65,-73.65),c(38.97,40.97))
grid_res=0.01
grid_crs="epsg:4326"
grid <- rast(nrows=diff(range(grid_bbox[,2]))/grid_res,
            ncols=diff(range(grid_bbox[,1]))/grid_res, xmin=min(grid_bbox[,1]),
```

```
        xmax=max(grid_bbox[,1]), ymin=min(grid_bbox[,2]), ymax=max(grid_bbox[,2]),
        crs=grid_crs)
grid_vect <- as.polygons(ext(grid),crs=grid_crs)

NWI_Wetland_fraction(input_directory=~/.//Desktop/in/",
                     output_directory=~/.//Desktop/out/",
                     Use_SOCCR1=TRUE,
                     Use_SOCCR2=TRUE,
                     Include_freshwater=TRUE,
                     domain=grid_vect,
                     domain_template=grid,
                     state_name_list=c("DE", "MD", "NJ", "NY", "PA"))
```

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